NATHANYL L. FREESE

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EDUCATION

Embry-Riddle Aeronautical University (ERAU) – Daytona Beach Campus **Exp. Grad:** December 2026
Bachelor of Science, Aerospace Engineering **GPA:** 3.97/4.00
Tau Beta Pi Engineering Honor Society

WORK EXPERIENCE

Collins Aerospace Internship, Melbourne, FL May 2026 – Present
System Engineering Intern

- Built Python automation scripts simulating keyboard/mouse input to execute 301 transponder test files, cutting runtime by 50%
- Evaluated an internal AI test case generation tool across 50+ requirements, providing and implementing feedback that improved accuracy and efficiency
- Authorized test cases and formal test procedures while tracking verification progress in JIRA and GitLab

KBR & LZ Technology Internship, NASA Johnson Space Center, TX January 2025 – August 2025
Engineering Intern

- Developed software tools using Python and PyQt5 to visualize antenna blockage, tracking, and Orion/Starliner simulations with live telemetry overlays for controller training and simulated missions
- Optimized the Gateway visualization tool achieving a 13x overall performance boost and 21x faster graphics rendering
- Led development of the Starliner visualization tool, delivering integrated telemetry overlays used in operational workflows

Thermodynamics Tutor, ERAU, FL January 2024 – January 2025

- Tutored over 400 students seeking help in Thermodynamics and Heat transfer each semester
- Created and provided targeted problem sets, exam reviews, and custom lesson plans to boost students' understanding

PROJECT EXPERIENCE

Mayott Aerospace Heavy Lift UAV's, Daytona Beach, FL April 2026 – Present
CAD & Software Engineer

- Modeling heavy-lift UAV frame components in Fusion360 and SolidWorks, optimizing for weight and structural efficiency
- Developing software in Rust, targeting memory safety and real-time performance

Crewed Lunar Mission Preliminary Design, ERAU, FL January 2026 – May 2026
Project & Systems Lead

- Led a 6-person team to design a crewed 2 stage LOX/CH4 lunar vehicle covering launched through splashdown
- Performed vehicle sizing in MATLAB and CEA and led trade studies across all subsystems to meet mission requirements
- Modeled full launch vehicle assembly in CATIA V.5 and Fusion360, producing detailed parts breakdown

Embry-Riddle Future Space Explorers and Development (ERFSEDS) Artemis Rocket, ERAU, FL August 2024 – January 2025
Manufacturing Member

- Developed 3-D models of a scaled-down version of the Artemis rocket using CATIA V.5 and SolidWorks
- Assisted in running simulations in OpenRocket to optimize trajectory, stability, and flight parameters
- Collaborated with a 15-student team to integrate design modifications and ensure structural integrity

Experimental Rocket Propulsion Lab (ERPL) Spectre Rocket, ERAU, FL August 2022 – May 2024
Hardware Member

- Assisted in a team of 10 students to create structural components of an active-stabilization high-powered rocket
- Optimized the rocket's design by soldering components and using Fusion360 to minimize aerodynamic forces
- Analyzed and documented test experiment data using MATLAB to drive continuous improvements

SKILLS

- CATIA V.5	- SolidWorks	- Fusion360	- Creo Parametric	- NASTRAN	- Ansys Fluid
- MATLAB	- Python	- Rust	- Programming C	- PyQt5 Designer	- Visual Studio
- GitLab / GitHub	- JIRA	- NASA CEA	- Microsoft Office	- Communication	- Linux Command Line